

Riza Mohammad Khan

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EDUCATION

National Institute of Technology

Bachelor of Technology in Computer Science

Nov. 2022 – July 2026

Srinagar, IN

EXPERIENCE

Software Developer

Automee

July 2024 – Jan 2025

Poland

- Designed and deployed AI agents, including voice bots, and CRM pipelines for 12 international clients using Python, Voiceflow, and JavaScript.
- Worked collaboratively in a geographically distributed team environment, coordinating development efforts with engineering professionals across three countries (Philippines, Poland, and India).

PROJECTS

le-merge | *TypeScript, Bun, React (Ink), Git, AI SDKs, Zod* | [le-merge](#)

June 2026

- Architected a terminal UI (TUI) application leveraging React-based Ink to resolve Git merge conflicts using autonomous AI agents equipped with local system tools and native SDK integrations (OpenAI, Anthropic, Gemini).
- Engineered a conservative line-range replacement mechanism that performs precise edits to eliminate complete-file rewrite hallucinations, paired with an interactive terminal dialog to resolve ambiguous conflict intents.
- Built a cross-platform compilation pipeline in Bun to output native binaries for macOS, Linux, and Windows, wrapped in a dynamic NPM packaging architecture that auto-detects host specifications during install.

Trix | *Rust, ALSA, Ratatui* | [trix-player](#)

Jan 2026

- Architected a terminal music player with direct ALSA integration for hardware-level audio control, achieving zero-latency playback and native system volume management with software fallback mechanisms.
- Engineered a high-performance Ratatui TUI supporting 60fps rendering, recursive directory indexing, sub-50ms fuzzy search, and continuous state controls (shuffle, loop modes, and timestamp seeking).
- Designed a deterministic TOML-based theming engine and established automated GitHub Actions CI/CD workflows to publish multi-target builds (.deb, .rpm, .tar.xz) to the Arch User Repository (AUR).

Serma | *Rust, Tantivy, Axum, DHT Protocol* | [serma-search](#)

Dec 2025

- Architected a decentralized BitTorrent DHT spider in Rust that implements BEP-5 and BEP-9 protocols to crawl and metadata-enrich 10k–50k torrents daily via asynchronous peer connections.
- Designed a high-throughput search system using Tantivy and Sled key-value storage, delivering sub-100ms full-text query response times across millions of indexed documents.
- Implemented an automated indexing garbage-collection model that filters low-quality entries, supported by Docker containerization, environment configuration, and SOCKS5 privacy-conscious proxy support.

COMMUNITY INVOLVEMENT

DreamDay Srinagar | *Hack Club Game Jam* | *Technical Mentor*

Sep 2025

- Served as a technical mentor at Kashmir Valley's first-ever game jam, supporting participants with game design, debugging, and rapid prototyping.

CodeDay Kashmir | *International Hackathon* | *Technical Mentor*

July 2025

- Mentored young developers at the region's inaugural international coding event, providing guidance on project ideation and programming fundamentals.

TECHNICAL SKILLS

Rust, C, Python, Bash, SQL, Git, Docker, Linux, Node, Javascript, Typescript